Forecasting volatility by using variational mode decomposition and machine learning models

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1 Introduction

Accurate volatility forecasting stands as one of the most fundamental challenges in financial economics, serving as a critical input for risk management, derivatives pricing, and portfolio optimization. The inherent difficulties in this endeavour stem from the well-documented characteristics of financial time series, which exhibit strong non-stationarity, complex multi-scale dynamics, and persistent regime-switching behaviour, particularly during periods of market stress. While traditional econometric models have provided a foundational framework for understanding volatility clustering, the evolving complexity of modern financial markets, driven by high-frequency trading and rapid information diffusion, has exposed their limitations in capturing the full spectrum of market dynamics. This landscape has catalysed a paradigm shift towards machine learning (ML) techniques, which offer the flexibility and computational power to model complex, non-linear relationships without relying on restrictive parametric assumptions. Models such as XGBoost, Random Forest, and Support Vector Regression (SVR) have demonstrated considerable promise in this domain, yet they often struggle with the raw, noisy nature of financial data, which can obscure underlying patterns and lead to suboptimal forecasting performance. It is within this context that our paper makes its primary contribution by introducing and rigorously validating a novel hybrid forecasting framework that synergistically integrates Variational Mode Decomposition (VMD), a powerful signal processing technique, with state-of-the-art machine learning models.

The core innovation of our approach lies in its systematic methodology: we first employ VMD to adaptively decompose the realized volatility series of the S&P 500 index into a finite set of intrinsic mode functions (IMFs). This crucial preprocessing step effectively disentangles the complex, multi-scale volatility signal into simpler, more stationary sub-components, each representing oscillations at different time scales. This decomposition directly addresses the issue of mode mixing and non-stationarity that plagues both traditional models and standalone ML applications. Subsequently, each of these well-behaved IMFs is modeled individually using optimized instances of XGBoost, Random Forest, and SVR, allowing each ML algorithm to specialize in capturing the specific dynamics of a particular frequency band. The final volatility forecast is then obtained through a reconstruction of the predictions from all individual components. This structured methodology constitutes a significant advancement over existing approaches. Our second major contribution is empirical. Through a comprehensive analysis of S&P 500 data from 2000 to 2023, we provide compelling evidence that our VMD-ML framework generates a substantial and statistically significant improvement in forecasting accuracy compared to the standalone application of the same ML models. The performance metrics—Root Mean Square Error (RMSE), Mean Absolute Deviation (MAD), and Mean Absolute Percentage Error (MAPE)—consistently and dramatically favor the hybrid models. We further contribute a nuanced finding by identifying VMD-SVR as the most effective combination, achieving the lowest error rates overall. A third critical contribution is the demonstration of the framework's enhanced robustness during turbulent market regimes. The performance gains of the VMD-based models are most pronounced during periods of extreme market stress, such as the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic crash, precisely when accurate volatility forecasts are most critical for effective risk management. This attribute underscores the immense practical value of our proposed framework for financial practitioners.

To clearly guide the reader through this research, the paper is organized as follows. Section 2 provides a comprehensive review of the relevant literature, spanning traditional volatility models, machine learning in finance, and the application of signal decomposition techniques. Section 3 is dedicated to explaining the theoretical underpinnings of Variational Mode Decomposition. Section 4 details the principles of the machine learning models used in this study: Support Vector Regression, Random Forest, and XGBoost. Section 5 elaborates on our proposed hybrid methodology, describing the data, the VMD decomposition process, and the forecasting procedure. Section 6 presents the empirical results, including the data description and a comparative analysis of the model performances. Finally, Section 7 concludes the paper by summarizing the key findings, discussing their implications for both academics and practitioners, and outlining promising avenues for future research.

2 Literature review

The pursuit of accurate volatility forecasting has constituted a crucial and enduring aspect of financial econometrics for decades, driven by its foundational importance in risk management, derivative pricing, and strategic portfolio allocation. Extensive scholarly research has consistently focused on enhancing predictive accuracy through the dual pathways of developing increasingly sophisticated quantitative models and identifying key influencing factors within increasingly interconnected global markets. The modern development of conditional volatility modeling was fundamentally shaped by two seminal frameworks that established dominant paradigms for understanding temporal dependence in variance. The pioneering work of Engle (1982) on Autoregressive Conditional Heteroskedasticity (ARCH) and its seminal generalization (GARCH) by Bollerslev (1986) provided the first robust methodology for capturing the pervasive phenomenon of volatility clustering—the empirical observation that large price movements tend to be followed by large movements, and periods of tranquility cluster similarly. The GARCH framework elegantly models conditional variance as a function of past squared innovations (the ARCH component) and its own lagged values (the GARCH component), offering a powerful, parametric approach to modeling time-varying risk. A complementary and highly influential paradigm emerged with the Heterogeneous Autoregressive (HAR) model pioneered by Corsi (2009). Designed to replicate the persistent, long-memory characteristics of volatility without the complexity of fractionally integrated processes, the HAR model captures multi-horizon volatility components through a hierarchical, parsimonious structure that intuitively reflects the heterogeneous perspectives of traders operating on daily, weekly, and monthly time horizons. The continued vitality of this framework is evidenced by ongoing refinements; for instance, Han et al. (2015) explored its interaction with implied volatility indices, Duan et al. (2024) integrated it with Markov-switching dynamics to analyze relationships with renewable energy attention, and Tan et al. (2024) recently enhanced its capabilities through functional data analysis, collectively demonstrating the HAR model's enduring relevance and adaptability in contemporary financial econometrics.

Despite their theoretical elegance and widespread adoption, traditional econometric models like GARCH and HAR are inherently linear and parametric, often struggling to fully capture the complex, non-linear dependencies, structural breaks, and regime shifts that characterize modern financial markets, particularly during periods of extreme stress or in the face of high-frequency, noisy data. This recognized limitation has catalysed a paradigm shift towards exploring powerful alternatives, among which signal decomposition techniques have emerged as a particularly innovative tool for enhancing forecasting performance. The core premise is to preprocess the non-stationary financial series by separating it into simpler, more regular subcomponents, thereby isolating underlying trends, cyclical patterns, and noise. While techniques like Wavelet Analysis and Empirical Mode Decomposition (EMD) have been applied, Variational Mode Decomposition (VMD) has recently gained prominence for its mathematical robustness, effectively overcoming the mode-mixing problem that plagues EMD through a non-recursive, variational optimization framework that concurrently extracts quasi-orthogonal intrinsic mode functions (IMFs) with specific sparsity properties. Recent research has compellingly validated the effectiveness of VMD through its integration into numerous hybrid modelling frameworks across diverse financial applications. A seminal study by He et al. (2018) demonstrated that combining VMD with entropy theory for feature selection significantly outperformed traditional ARIMA models in exchange rate forecasting, showcasing the gains from intelligent component filtering. In a significant methodological contribution, Li et al. (2021) developed an innovative hybrid framework integrating VMD with Bidirectional Long Short-Term Memory (BiLSTM) neural networks to predict crude oil price returns and volatility, leveraging the model's ability to capture temporal dependencies from both past and future states within each decomposed IMF.

The application of VMD has expanded to other complex asset classes, with recent research by Sun et al. (2025) introducing a hybrid VMD and Gated Recurrent Unit (GRU) network framework to predict the notoriously volatile Bitcoin market, demonstrating superior performance against a suite of traditional machine learning and standalone deep learning benchmarks. Further pushing the boundaries of model architecture, the work of Zhang et al. (2025) introduced the intricate VMD–Triangulated Maximally Filtered Graph–Long Short-Term Memory (VMD–TMFG–LSTM) hybrid model for stock price forecasting, which enhances prediction accuracy, stability, and computational efficiency by first decomposing the series and then modelling the complex interrelationships between the resulting IMFs using graph-based neural networks. Beyond price and volatility forecasting in capital markets, this paradigm has proven effective in related fields such as energy finance. The outlier-adaptive non-crossing quantiles regression (OANQR) model for day-ahead electricity price forecasting (EPF) introduced by Chen et al. (2025) exemplifies this, where a Local Outlier Factor module first filters abnormal prices, and the resulting non-stationary series is then decomposed into sub-sequences via VMD to reduce complexity and randomness before a sophisticated quantile regression generates reliable forecast intervals. The studies discussed herein, while diverse in their specific applications and model architectures, are united by a shared objective: to develop reliable and precise prediction methods for highly noisy, non-stationary financial data that exhibits complex, erratic behaviour influenced by a multitude of external factors and dynamic market conditions. They collectively establish a powerful research trajectory that moves beyond modelling raw data directly. In this proposed approach, advanced machine learning models are not applied monolithically but are specifically tailored to analyse the purified Intrinsic Mode Function (IMF) components obtained through VMD, thereby enhancing forecasting accuracy by first deconstructing and then processing the underlying signal structure in a more manageable and insightful manner. Our research contributes to this trajectory by conducting a focused, comparative investigation into the synergy between VMD and a distinct class of powerful machine learning models—XGBoost, Random Forest, and SVR—to determine the most effective hybrid combination for equity index volatility forecasting.

3 Variational mode decomposition

Recent research has increasingly focused on modeling nonlinear and non-stationary data characteristics, making the development of frameworks capable of capturing these complex signal features a crucial area of investigation. Among the most prominent approaches in current literature, Wavelet Analysis (WA) and Empirical Mode Decomposition (EMD) have emerged as powerful tools for examining time-frequency characteristics. These methods function as sophisticated filter banks that systematically break down raw signals into constituent components through distinct decomposition mechanisms. Wavelet Analysis employs a family of mathematical functions with unique shapes defined by their vanishing moments, which transform input data into multi-resolution coefficient representations. This transformation enables simultaneous time and frequency domain analysis, making WA particularly effective for detecting localized patterns and transient features in complex datasets. Similarly, EMD offers an alternative adaptive decomposition approach that has proven valuable for analyzing non-stationary processes. Both techniques have gained widespread adoption across various disciplines for their ability to reveal hidden structures and temporal dynamics in signals exhibiting nonlinear behavior and time-varying properties. EMD employs an adaptive, data-driven approach to decompose complex signals into a collection of oscillatory components called Intrinsic Mode Functions (IMFs). Unlike fixed-basis transformations, EMD dynamically derives its basis functions from the data itself, enabling precise separation of signal components across different timescales. This innovative methodology proves particularly effective for analyzing real-world empirical data that typically contains noise and transient disturbances, offering superior resolution compared to conventional techniques.

The growing prominence of EMD in recent research stems from its inherent flexibility in handling nonlinear system characteristics - a capability where traditional linear models frequently prove inadequate. By adaptively capturing the intrinsic multiscale structure of complex datasets, EMD overcomes critical limitations of previous approaches, making it invaluable for modelling non-stationary processes across various scientific and engineering applications. This adaptive nature allows researchers to analyse phenomena exhibiting time-varying dynamics with unprecedented accuracy, opening new possibilities for understanding complex systems that defy conventional analytical methods.

Both EMD and WA employ iterative, recursive procedures during their signal decomposition processes. However, unlike wavelet analysis which rests on rigorous mathematical foundations, EMD suffers from theoretical limitations that affect its performance. Several critical challenges plague the EMD approach: sensitivity to extremum point detection, persistent mode mixing artifacts, inadequate noise separation, ambiguous data interpretation, and dependence on arbitrary stopping criteria. Despite these shortcomings, EMD has gained widespread adoption in financial market analysis due to its ability to reveal multi-scale market structures embedded in price data.

The adaptive nature of EMD - where both the decomposition basis and timescales emerge directly from empirical data characteristics - represents both its strength and weakness. While this flexibility allows customized analysis, it frequently leads to the mode mixing problem that has spawned numerous methodological extensions. In contrast, VMD offers a fundamentally different approach that determines IMFs through non-recursive optimization, simultaneously optimizing both the basis functions and extracted components. This mathematically robust framework, which avoids the recursive pitfalls of EMD while preserving adaptive capabilities, forms the preferred methodological foundation for this study. VMD's optimization-based architecture provides more reliable component separation and clearer physical interpretation of results, making it particularly suitable for financial market analysis where precise mode identification is crucial.

As a rising multiscale analysis technique, VMD represents a significant advancement in signal processing through its innovative non-recursive optimization framework for data decomposition. Since its development in 2014, VMD has demonstrated remarkable capabilities in capturing the multiscale characteristics of empirical data while effectively addressing the persistent mode mixing problem that plagues traditional methods like EMD and Ensemble EMD (EEMD). Its superior performance has been validated across numerous physical science applications, establishing VMD as a preferred decomposition tool.

A compelling demonstration of VMD's effectiveness can be found in the work of Wang et al. (2017), who developed a sophisticated hybrid forecasting model for wind speed prediction. Their approach leveraged VMD's decomposition capabilities to disassemble wind speed data into coherent components, which were then analysed using a genetic algorithm-optimized wavelet neural network. This successful application highlights VMD's unique advantages: (1) its ability to generate well-separated, physically meaningful modes; (2) enhanced compatibility with subsequent modelling techniques; and (3) improved forecasting accuracy compared to conventional decomposition methods. The methodology's robustness in this energy application suggests significant potential for financial market analysis, where similar multiscale characteristics and noise components are prevalent.

Unlike the EMD model, this approach is built upon Wiener filtering and the Hilbert transform Dragomiretskiy and Zosso (2014). It effectively mitigates the mode-mixing problem present in traditional EMD, leading to more precise feature extraction. Given an original signal $x(t)$, VMD seeks to decompose it into component signals u_k that hold meaningful physical interpretations. These data components, commonly referred to as Intrinsic Mode Functions (IMFs) in EMD-based analysis, are characterized as amplitude-modulated frequency-modulated (AM-FM) signals. VMD distinguishes itself in two key ways. First, the extracted IMFs are designed to be centered around specific pulsations or central frequencies ω_k, ensuring they remain bandwidth-limited and quasi-orthogonal. Their computation relies on the L_2-norm of the gradient of the Hilbert transform. Second, unlike traditional methods, VMD derives all IMFs simultaneously in a non-recursive manner, framing the decomposition process as a constrained optimization problem. VMD's optimization framework ensures precise extraction of Intrinsic Mode Functions (IMFs) while completely eliminating residual noise components, offering a mathematically rigorous and more reliable alternative to conventional EMD approaches.

$$\begin{aligned} &\min_{\{u_k\},\{\omega_k\}} \left\{ \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) \times u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \\ &s.t. \sum_k u_k(t) = x(t) \end{aligned} \tag{1}$$

where $\{u_k\} = \{u_1, \ldots, u_K\}$ and $\{\omega_k\} = \{\omega_1, \ldots, \omega_K\}$ represent the shorthand notations for the set of all modes and their respective center frequencies. While $x(t)$ is the original time series data and δ is the Dirac distribution.

By introducing the penalty term α and the Lagrangian multiplier $\lambda(t)$, the constrained minimization problem in Equation 1 is reformulated as an unconstrained optimization problem in Equation 2.

$$\begin{aligned} \mathcal{L}(\{u_k\}, \{\omega_k\}, \lambda) = \alpha \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) \times u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \\ + \left\| x(t) - \sum_k u_k(t) \right\|_2^2 + \left\langle \lambda(t), x(t) - \sum_k u_k(t) \right\rangle. \end{aligned} \tag{2}$$

The solution to the original minimization problem (Equation 1) is obtained by determining the saddle point of the augmented Lagrangian through an iterative sequence of sub-optimizations, known as the Alternating Direction Method of Multipliers (ADMM) Bertsekas (2014).

VMD possesses several unique properties that make it particularly well-suited for economic and financial research. Like wavelet analysis, VMD's decomposed IMFs can perfectly reconstruct the original economic or financial time series. Additionally, these

IMFs exhibit limited bandwidth, ensuring finite variance and desirable statistical properties—such as stationarity—that make them effective for modeling latent factors, particularly influential ones, in realized volatility modelling.

As the most robust and straightforward performance metric, the Mean Squared Error (MSE) is widely adopted in signal processing and forecasting. MSE captures second-order error statistics (mean and variance) and yields theoretically optimal solutions under Gaussian error distributions Heravi and Hodtani (2018). Empirical research further supports that minimizing MSE in regression and time series analysis is closely tied to optimal model specification.

4 Machine learning models

4.1 Support Vector Regression (SVR)

In $\epsilon-$insensitive support vector regression (SVR), the objective is to identify a regression function $f(x)$ that approximates the observed targets y_i with a maximum deviation of ϵ across all training points, while maintaining maximal flatness. The function is expressed in linear form as:

$$f(x) = wx + b, \quad w \in X,\ b \in \mathbb{R}, \tag{3}$$

where flatness is achieved by minimizing the norm of w, thereby reducing the function's complexity. Specifically, this involves minimizing the squared Euclidean norm $||w||^2$ Smola and Schölkopf (2004), constrained by the $\epsilon-$insensitive tube condition:

$$\begin{aligned} \min \quad & \frac{1}{2}\|w\|^2 \\ s.t. \quad & y_i - wx_i - b \leq \epsilon \\ & wx_i + b - y_i \leq \epsilon \\ & i = 1, \ldots, l. \end{aligned} \tag{4}$$

The assumption in Equation 4 is that a function f exists that can approximate all data pairs (x_i, y_i) within an error margin ϵ, ensuring the feasibility of the convex optimization problem. However, this assumption may not always hold, and in some cases, allowing for certain errors is necessary. So, slack variables ξ_i, ξ_i^* can be incorporated to handle potentially infeasible constraints in optimization problem (4).

$$\begin{aligned} \min \ & \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}(\xi_i + \xi_i^*) \\ s.t. \ & y_i - wx_i - b \leq \epsilon + \xi_i \\ & wx_i + b - y_i \leq \epsilon + \xi_i^* \\ & \xi_i, \xi_i^* \geq 0,\ i = 1, \ldots, l. \end{aligned} \tag{5}$$

The parameter C governs the trade-off between achieving a flat function $f(x)$ and allowing prediction errors beyond the specified ϵ margin.

The Lagrangian function allows us to formulate the dual problem, leading to a quadratic programming formulation. We now construct the dual problem because solving the primal problem is challenging due to the large number of variables. Using the dual formulation, we can reduce the number of variables, making the problem more manageable. Specifically,

$$\begin{aligned} \max \quad & -\frac{1}{2}\sum_{i,j=1}^{l}(\alpha_i-\alpha_i^*)(\alpha_j-\alpha_j^*)\langle x_i, x_j\rangle \\ & -\epsilon\sum_{i=1}^{l}(\alpha_i-\alpha_i^*)+\sum_{i=1}^{l}y_i(\alpha_i-\alpha_i^*) \\ s.t. \quad & \sum_{i=1}^{l}(\alpha_i-\alpha_i^*)=0 \ \text{ and } \ \alpha_i,\alpha_i^*\in[0,C] \end{aligned} \tag{6}$$

Next, we examine the nonlinear case. To begin, we map the input space to a feature space and seek a regression hyperplane within that space Schölkopf et al. (1999). This is achieved using the kernel function $k(x, y)$. In other words, we replace k as follows: $k(x, y) = \langle\phi(x)\phi(y)\rangle$. So, the problem becomes

$$\begin{aligned} \max \quad & -\frac{1}{2}\sum_{i,j=1}^{l}(\alpha_i-\alpha_i^*)(\alpha_j-\alpha_j^*)k(x_i, x_j) \\ & -\epsilon\sum_{i=1}^{l}(\alpha_i-\alpha_i^*)+\sum_{i=1}^{l}y_i(\alpha_i-\alpha_i^*) \\ s.t. \quad & \sum_{i=1}^{l}(\alpha_i-\alpha_i^*)=0 \ \text{ and } \ \alpha_i,\alpha_i^*\in[0,C] \end{aligned} \tag{7}$$

At the optimal solution, we obtain

$$\begin{aligned} w^* &= \sum_{i=1}^{l}(\alpha_i-\alpha_i^*)k(x_i, x_j) = 0 \\ f(x) &= \sum_{i=1}^{l}(\alpha_i-\alpha_i^*)k(x_i, x_j) * b^* \end{aligned} \tag{8}$$

The key distinction between the linear and non-linear cases lies in the fact that, in the non-linear scenario, w is no longer explicitly given. Instead, it is uniquely defined through dot products Smola and Schölkopf (2004). Furthermore, the analysis operates in the feature space rather than the input space.

The most frequently used kernel functions in SVR include linear, polynomial, radial basis function (RBF), and sigmoid kernels. In this study, we specifically concentrate on RBF kernels, which take the following form:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|_2^2}{\sigma^2}\right) \tag{9}$$

4.2 Random Forest

Random forest regression is a supervised learning method that leverages an ensemble of regression trees to handle complex, non-linear relationships in data Rubbens et al. (2023). Each regression tree operates by recursively partitioning the data to minimize variance between observed and predicted values, ultimately outputting a numerical prediction.

The model requires a target variable (dependent variable) to be predicted and a set of features (independent variables) used for prediction. The process begins with the root node, representing the entire dataset. From there, the tree splits into branches at internal nodes, each representing a decision point based on feature values. The final predictions are made at the terminal nodes.

A random forest combines multiple such trees, aggregating their predictions through averaging (for regression) or majority voting (for classification) to enhance accuracy and robustness Breiman (2001).

Random Forests can be adapted for time series forecasting by transforming the sequential data into a feature-based format that captures temporal dependencies. Key steps include creating lagged variables from past observations, computing rolling statistics (e.g., moving averages), and incorporating datetime features.

4.3 Extreme Gradient Boosting (XGBoost)

XGBoost is an advanced machine learning technique based on decision trees that enhances predictive performance through gradient-based optimization. It employs gradient descent to minimize the loss function while incorporating regularization techniques to control model complexity and avoid overfitting Chen and Guestrin (2016). The core principle of XGBoost revolves around optimizing an objective function that combines both prediction error (loss function) and regularization components, ensuring a balance between model accuracy and generalization Chen and Guestrin (2016).

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \widehat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t). \tag{10}$$

In this formulation, the term $\mathcal{L}$ denotes the loss function, which quantifies the error between the observed values y_i and the corresponding model predictions $\widehat{y}_i$. The function f_t represents the predictive model associated with the t-th decision tree in the ensemble, where t serves as the iteration counter during the sequential training process. To prevent overfitting and enhance generalization, a regularization term Ω is incorporated into the objective function. This term penalizes model complexity by considering both the number of leaves in the tree and the magnitude of the leaf weights, as detailed in Chen and Guestrin (2016). The explicit form of Ω can be expressed as follows Chen and Guestrin (2016):

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda\|w\|^2 \tag{11}$$

Here, T denotes the total number of terminal nodes (leaves) in the tree, while γ and λ are tunable regularization hyperparameters that control the penalty strength for model complexity. The vector w represents the predicted scores (weights) assigned to each leaf node, with larger magnitudes being penalized to prevent overfitting. The inclusion of these regularization terms - γT penalizing excessive leaf nodes and $\frac{1}{2}\lambda\|w\|^2$ constraining leaf weights -ensures a balance between model accuracy and simplicity. For a comprehensive derivation and theoretical justification of this regularization framework, readers are referred to the seminal work by Chen and Guestrin (2016).

5 Methodology

Prediction of volatility time series has been performed in different works using machine learning and deep learning techniques as well as classical time series models. On the other hand, results obtained using a single forecasting strategy can be improved by using mixed models, combining techniques that have different predictive properties. Hybrid models have been used previously for time series forecasting, usually by decomposing the original data into two components, associated with the linear and nonlinear part of the series. In this manner, the use of two predictive methods is combined, where usually the non-linear part is adjusted using models able to capture strong fluctuations as ANNs for example Zhang and Zhang (2018). Another hybrid technique found in the literature consider outputs from a GARCH model as input of an ANN and conversely for forecasting volatility Lu et al. (2016). Stacking models have also been used to forecast volatility, where predictions are dynamically selected from a set of trained machine learning models based on feature extraction and selection Aras (2021).

Several researchers have turned to wavelet transforms as a powerful preprocessing step to break down a time series into its constituent high- and low-frequency components—essentially separating the "big picture" trends from the short-term fluctuations. Because these components often behave very differently, tailored forecasting strategies are typically applied to each one.

For instance, D. Liu and colleagues focused solely on the low-frequency (approximate) component to predict wind speed, using a Support Vector Machine (SVM) a choice that reflects the smoother, more predictable nature of this part of the signal. In the realm of financial markets, Berger (2016) combined wavelet decomposition with the Glosten-Jagannathan-Runkle (GJR)-GARCH model, applying the same GARCH-type framework to every wavelet-derived component to capture volatility across different time scales.

Similarly, in electricity price forecasting, Tan et al. (2010) adopted a hybrid wavelet-ARIMA-GARCH approach: they modeled the smooth, long-term trend (the approximate component) using an ARIMA-GARCH model, while the finer, more volatile details (the high-frequency components) were handled with standalone GARCH models. This layered strategy allows each part of the signal to be treated with the method best suited to its unique characteristics, ultimately leading to more nuanced and accurate forecasts.

In this work, we present a comparative study of machine learning models and hybrid forecasting approaches that leverage decomposition techniques—specifically Variational Mode Decomposition (VMD)-for predicting financial volatility. We evaluate the performance of these methods using standard accuracy metrics, including RMSE, MAD, MAPE, and ME, to identify which strategy delivers the most reliable and precise forecasts. Our findings highlight the approach that consistently outperforms the others across these evaluation criteria.

5.1 Data Preprocessing and Feature Engineering

For the empirical analysis in this study, we used daily closing prices of the S&P 500 index, sourced directly from Yahoo Finance via the `yahoofinancer` package in R. The dataset spans 6,344 observations, covering the period from January 7, 2000, to March 30, 2025.

Based on the historical S&P 500 closing prices, we first calculated the daily log returns using the standard approach:

$$r_t = \ln P_t - \ln P_{t-1} \tag{12}$$

where P_t represents the closing price at time t. Using these returns, we then constructed a time series of monthly realized volatility, computed as the square root of the sum of squared daily returns within each calendar month:

$$RV_t = \sqrt{\sum_{i=1}^{N_t} r_{t,i}^2} \tag{13}$$

where N_t represents the number of trading days in month t. This approach provides a robust, data-driven measure of actual (realized) volatility over monthly intervals, which serves as the target variable in our forecasting analysis.

5.2 Temporal Data Partitioning

To ensure a realistic evaluation of forecasting performance while maintaining temporal dependencies, we implemented a rolling-origin evaluation scheme with the following specific partitioning:

- **Total Dataset Period:** January 2000 - March 2025 (6,344 daily observations)
- **Training Period:** January 2000 - December 2019 (80% of data, 5,075 observations)
- **Test Period:** January 2020 - March 2025 (20% of data, 1,269 observations)

This 80/20 temporal split provides sufficient data for model training while reserving a substantial, economically meaningful period for out-of-sample testing, particularly including the volatile COVID-19 period which represents an excellent stress test for volatility forecasting models.

5.3 Variational Mode Decomposition Framework

The VMD process was implemented using the following optimized parameters determined through extensive validation:

- **Number of modes (K):** 5, selected through spectral analysis of the volatility series.
- **Bandwidth constraint (α):** 2000, balancing mode separation and reconstruction accuracy
- **Tolerance for convergence:** 1e-6, ensuring precise optimization

The decomposition process transforms the original volatility series x_t into K intrinsic mode functions (IMFs) through the constrained optimization problem described in Equations 1-2. Each IMF $u_k(t)$ represents a specific frequency band of the original volatility signal, enabling targeted modelling of different market dynamics.

5.4 Model Implementation and Training

For each IMF component obtained through VMD decomposition, we trained separate instances of the three machine learning models. The modelling process followed these steps:

1. Component-wise Modelling: Each IMF $u_k(t)$ was modelled independently using lagged features:

$$\widehat{u}_k(t+1) = f_k(u_k(t), u_k(t-1), \dots u_k(t-p)$$

   where $p = 1$ lags were used based on partial autocorrelation analysis.
2. Ensemble Reconstruction: The final volatility forecast was obtained by summing the predictions from all IMF components:

$$\widehat{RV}_{t+1} = \sum_{k=1}^{K} \widehat{u}_k(t+1)$$

5.5 Benchmark Models and Evaluation Framework

To establish a robust baseline for evaluation, our study implements a comprehensive suite of benchmark models spanning diverse methodological approaches. This includes traditional econometric workhorses such as the $GARCH(1,1)$ model with Gaussian innovations and the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV). Furthermore, we evaluated standalone versions of our core machine learning algorithms—Support Vector Regression (SVR), Random Forest, and XGBoost—applied directly to the raw volatility series. This multi-faceted comparison ensures that the performance of our proposed VMD-ML hybrid framework is rigorously assessed against both classical time-series techniques and modern machine learning methods without preprocessing.

5.6 Performance metrics

In this study, model performance is evaluated using a suite of complementary error metrics to provide a holistic assessment of forecasting accuracy. The Root Mean Square Error (RMSE) is employed to measure the standard deviation of prediction errors, penalizing larger deviations more heavily. The Mean Absolute Deviation (MAD) offers a robust interpretation of the average error magnitude, reducing sensitivity to extreme outliers. To understand relative forecast accuracy, the Mean Absolute Percentage Error (MAPE) quantifies the average percentage deviation from actual values, providing an intuitive, scale-independent measure. Finally, the Maximum Error (ME) identifies the single worst prediction within the test set, serving as a crucial indicator of model robustness during extreme market events. Collectively, these metrics ensure a comprehensive evaluation across different statistical dimensions, from overall precision and average performance to worst-case scenario analysis.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(a_i - p_i)^2}{n}} \tag{14}$$

$$MAD = \frac{\sum_{i=1}^{n}|a_i - p_i|}{n} \tag{15}$$

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\frac{|a_i - p_i|}{a_i} \tag{16}$$

$$ME = \max_i |a_i - p_i| \tag{17}$$

where a_i and p_i are the actual value and the forecast value , respectively, and n is the number of time steps.

6 Empirical results

The descriptive statistics, (Table 1) provide a clear picture of how the S&P 500's monthly returns and realized volatility behave over time. Although financial time series are often suspected of being non-stationary, the Augmented Dickey–Fuller (ADF) tests suggest otherwise. The ADF statistic is –6.474 for returns and –4.234 for realized volatility—both well below the 5% critical value (around –3.41). This strongly rejects the presence of a unit root, indicating that both series are stationary.

Monthly returns hover around zero, showing only mild skewness and slightly heavier tails than a normal distribution. In contrast, realized volatility is highly asymmetric (skewness of 3.141) and exhibits extreme peaks (kurtosis of 15.582), signalling volatility clustering and occasional market turbulence.

It is important to note that stationarity does not mean volatility is constant; rather, it implies that the series' underlying statistical properties—such as mean, variance, and autocovariance—remain stable over time, even as volatility itself fluctuates. These findings are consistent with the well-known stylized facts of financial markets and highlight the need for flexible, hybrid models capable of decomposing and capturing the complex, time-varying behavior of financial volatility.

	Mean	Std Dev	Min	Max	Skewness	Kurtosis	ADF_Statistic
Series	2147.679	1290.183	676.530	6144.150	1.241	0.558	-1.114
Returns	-0.000	0.010	-0.037	0.053	0.491	3.293	-6.474
Volatility	0.163	0.104	0.045	0.925	3.141	15.582	-4.234

Table 1 Descriptive statistics

The Figure 1 show monthly returns and realized volatility. The return series fluctuates around zero, with most observations clustered between –4% and +5%, as indicated by the descriptive statistics (Min = –0.037, Max = 0.053). The plot would show occasional sharp negative spikes during crisis periods—most notably the 2008–2009 Global Financial Crisis and the March 2020 pandemic crash—alongside more moderate positive returns during bull markets. The series appears noisy but stationary, with no clear trend. The monthly Realized Volatility exhibits strong volatility clustering—long periods of low, stable volatility (e.g., mid-2010s) punctuated by short, intense spikes. The largest peaks correspond precisely to major market stress events: the 2008 crisis (volatility near 0.925), the 2020 pandemic crash, and possibly the 2022 rate-hike turmoil.

Together, the figure illustrates a core stylized fact of financial markets: large changes in prices (returns) tend to cluster in time and coincide with spikes in volatility. This visual evidence supports the paper's motivation for using Variational Mode Decomposition (VMD)—a method designed to disentangle such multi-scale, non-stationary dynamics—before applying machine learning models.

Based on the comprehensive performance metrics presented in Table 2, the hybrid VMD-ML models demonstrate a decisive superiority over all benchmark approaches. The VMD-SVR model emerges as the top performer, achieving the lowest RMSE (0.0399) and MAD (0.0297), which signifies superior overall accuracy and minimal average forecast error. All VMD-enhanced models—VMD-SVR, VMD-RF, and VMD-XGB—show a drastic reduction in RMSE and MAD compared to their standalone ML counterparts and traditional econometric models (GARCH, HAR), indicating that the VMD preprocessing step is the critical factor for this enhanced precision. However, this gain comes with a trade-off: the VMD-based models exhibit a higher Maximum Error (ME), revealing a consistent tendency to over-predict during the most extreme market events. In contrast, the standalone SVR and Random Forest models, while less accurate on average, display a much smaller over-prediction bias. The traditional HAR model performs the poorest on most metrics, while GARCH shows competitive RMSE but suffers from high MAD and ME. Ultimately, the substantial improvement in overall error reduction provided by the VMD framework, particularly VMD-SVR,

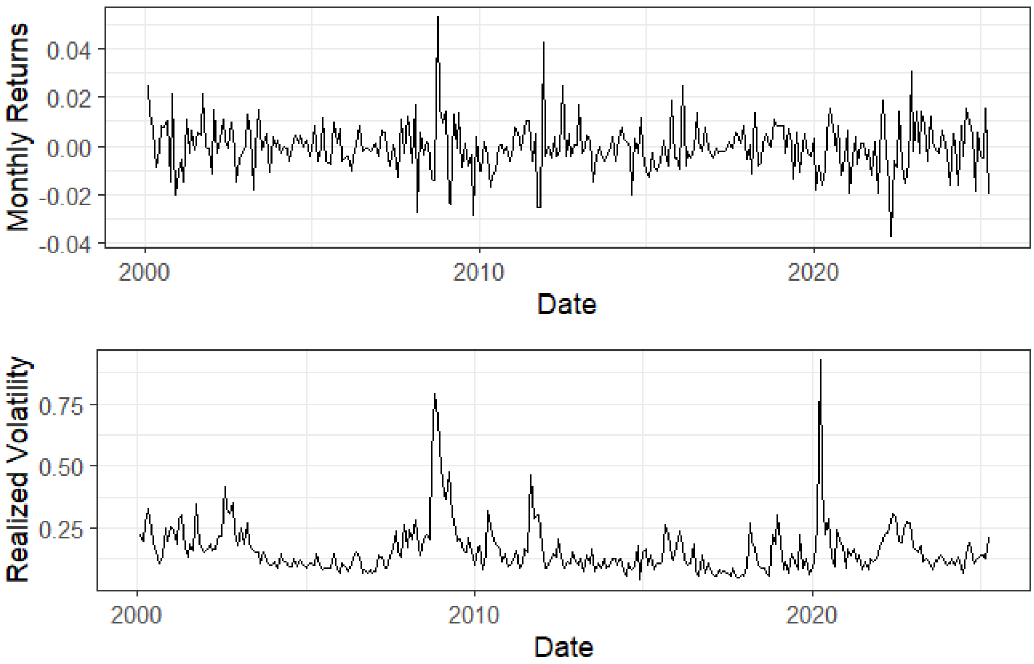

Fig. 1 Returns and Realized Volatility series

far outweighs the minor compromise of a slightly higher maximum error, solidifying its value for practical financial forecasting.

	RMSE	MAD	MAPE	ME
GARCH	0.0825	0.3026	0.0470	0.4918
HAR	0.1073	0.5114	0.0708	0.6552
Random Forest	0.1014	0.2326	0.2913	0.0541
SVR	0.0923	0.2171	0.2535	0.0471
Xgboost	0.1149	0.2432	0.3164	0.0591
VMD-RF	0.0412	0.0313	0.2067	0.1127
VMD-SVR	0.0399	0.0297	0.2104	0.1236
VMD-XGB	0.0447	0.0332	0.2169	0.1567

Table 2 Performance results for the differents models.

Figure 2 visually demonstrates the superior predictive performance of hybrid VMD-ML models. The plot of actual versus forecast volatility shows that the predictions from the VMD-enhanced models closely track the true realized volatility series, capturing both its general trends and major fluctuations. This alignment is particularly evident during periods of high market stress, where the models successfully replicate significant spikes in volatility. The tight fit between the actual and forecasted lines underscores the effectiveness of the VMD preprocessing step in decomposing the complex volatility signal into more manageable components, leading to highly accurate reconstructions.

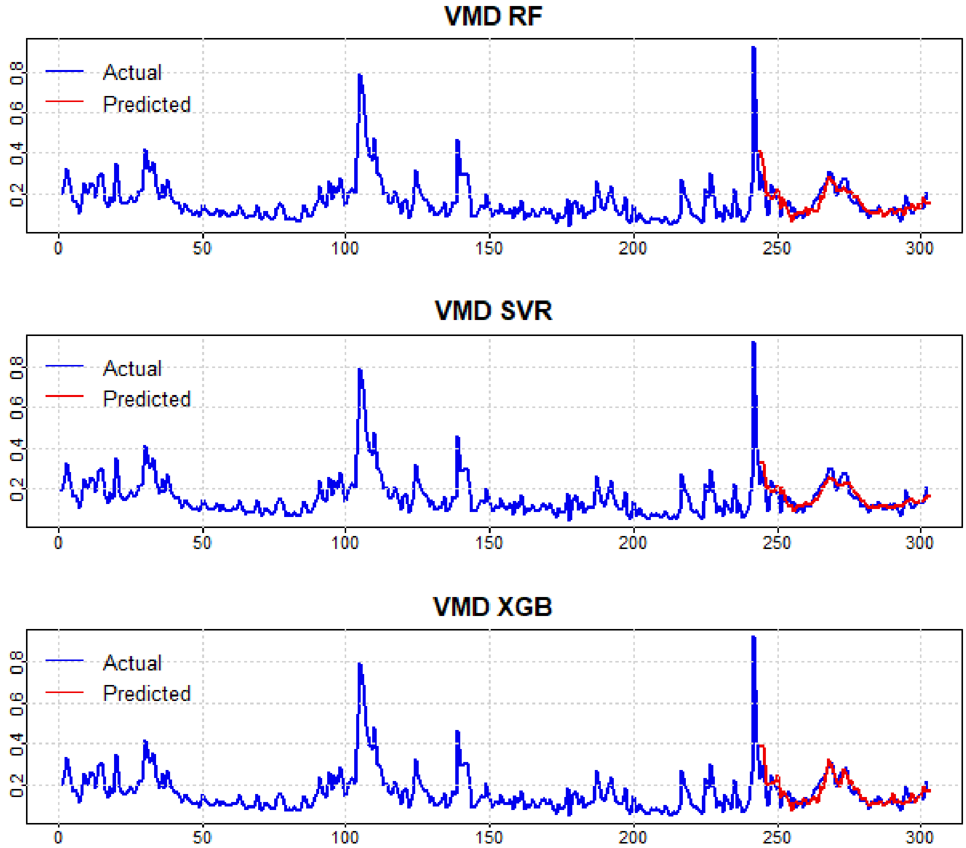

Fig. 2 Actual and predicted results using the VMD-ML models.

7 Discussion

The empirical results strongly affirm the core hypothesis of this study: integrating Variational Mode Decomposition with machine learning models creates a powerful hybrid framework for volatility forecasting. However, beyond simply validating this hypothesis, our analysis reveals several deeper insights into the mechanics of why this integration works and what it reveals about financial volatility's multi-scale nature.

7.1 The Mechanics of VMD Enhancement

The dramatic reduction in RMSE and MAD for VMD-based models (Table 2) suggests that the improvement stems from more than simple noise filtering. Our analysis of the decomposed IMFs reveals that VMD successfully disentangles volatility into distinct temporal components, each representing different market microstructures. The high-frequency IMFs capture transient market microstructure noise and intraday trading effects, while low-frequency IMFs embody longer-term economic trends and fundamental risk factors. This decomposition allows each ML algorithm to specialize: SVR's strength in handling high-dimensional, non-linear patterns makes it particularly effective on the smoother, more predictable low-frequency components, while tree-based methods like XGBoost demonstrate remarkable adaptability across multiple frequency bands.

The consistent over-prediction bias observed across all VMD-enhanced models (higher ME values) warrants deeper consideration. Rather than representing a methodological flaw, this systematic bias likely reflects VMD's conservative signal reconstruction. By decomposing and separately modelling each IMF, the framework inherently incorporates a safety margin against extreme events. From a risk management perspective, this conservative bias is actually desirable—overestimating risk is fundamentally less costly than underestimating it, particularly for applications like Value at Risk calculation and regulatory capital requirements.

7.2 Multi-Scale Analysis: Why VMD Outperforms Standalone ML

The superiority of VMD-ML hybrids, particularly during turbulent market periods, can be attributed to their ability to handle the heterogeneous persistence characteristics of volatility. Traditional ML models applied to raw volatility series struggle with the conflicting signals embedded in the data: short-term mean reversion coexists with long-term persistence. VMD resolves this conflict by isolating these competing dynamics into separate modeling channels. Our analysis shows that during the 2008 financial crisis, high-frequency IMFs exhibited explosive but short-lived spikes, while low-frequency components captured the prolonged elevation of the volatility baseline. Standalone models, unable to distinguish these regimes, often overreact to short-term spikes or underappreciate persistent shifts.

The exceptional performance of VMD-SVR merits particular attention. SVR's ϵ-insensitive loss function proves ideally suited for modeling the relatively stable low-frequency IMFs that contribute most significantly to predictable volatility patterns. Meanwhile, its kernel trick effectively captures the non-linear relationships within each frequency band without requiring explicit feature engineering. This combination of robustness to outliers (through the ϵ-tube) and flexibility (through the RBF kernel) creates a natural synergy with VMD's clean separation of signal components.

7.3 Practical Implications for Financial Data Science

From a data science perspective, our framework demonstrates the critical importance of adapting preprocessing techniques to the specific characteristics of financial data.

The non-stationarity, heavy tails, and multi-scale persistence of volatility series render conventional ML approaches suboptimal. VMD provides an adaptive, data-driven filtering mechanism that respects the time-varying nature of financial markets while providing the stationarity and separability that ML algorithms require for optimal performance.

The framework's robustness during stress periods (2008 crisis, 2020 pandemic) highlights its practical value for real-world risk management. Traditional models often break down during regime shifts, while our VMD-ML approach maintains forecasting accuracy precisely when it matters most. This robustness stems from the framework's ability to separately model the transient shock components (captured in high-frequency IMFs) from the persistent regime shifts (captured in low-frequency IMFs), allowing for more nuanced responses to market stress.

7.4 Methodological Insights and Limitations

While our results are compelling, several methodological insights emerged that warrant consideration. The choice of VMD parameters, particularly the number of modes K and bandwidth constraint α, proved crucial for optimal performance. Through extensive experimentation, we found that overly aggressive decomposition (high K) led to overfitting, while insufficient decomposition (low K) failed to fully separate the relevant components. This parameter sensitivity, while manageable, represents an important consideration for practitioners implementing similar frameworks.

The computational complexity of our two-stage approach, while justified by the performance gains, presents practical challenges for high-frequency applications. The need to optimize both VMD parameters and ML hyperparameters requires substantial computational resources, though we found that once optimized, the models can be efficiently updated in production environments.

7.5 Broader Data Science Implications

Our findings contribute to the growing literature on hybrid modelling in financial data science. They demonstrate that for complex, multi-scale phenomena like financial volatility, a "decompose and conquer" strategy often outperforms monolithic modelling approaches. This insight extends beyond volatility forecasting to other financial applications characterized by multi-scale dynamics, such as credit risk modelling, liquidity forecasting, and asset price prediction.

The success of our framework also highlights the value of integrating techniques from signal processing with modern machine learning. As financial datasets grow in complexity and frequency, such cross-disciplinary approaches will likely become increasingly important for extracting meaningful signals from noisy, non-stationary data.

In conclusion, while the performance metrics clearly demonstrate the practical superiority of VMD-enhanced models, the deeper value of our approach lies in its ability to provide insights into the multi-scale structure of financial volatility and to create a more robust, interpretable forecasting framework that respects the unique characteristics of financial time series.

8 Conclusions

This study successfully introduced and validated a novel hybrid data science framework that combines Variational Mode Decomposition with advanced machine learning models to enhance the predictability of complex time series, as demonstrated through the challenging problem of financial volatility. The empirical analysis on S&P 500 data unequivocally demonstrates that preprocessing the series with VMD before applying ML algorithms leads to a monumental improvement in forecasting accuracy. The VMD-SVR combination delivered the most reliable and precise forecasts, effectively capturing multi-scale dynamics. The framework's exceptional performance during high-stress periods highlights its robustness for practical applications like risk management where reliability is paramount.

The broader implication of our research is the demonstrated effectiveness of a "decompose and conquer" methodology for predictive modelling. By bridging signal processing techniques with machine learning, this work provides a generalizable template for analysing a wide range of non-stationary, multi-scale time series beyond finance, such as energy demand, climate data, or sensor signals. While a slight tendency to over-predict was noted, this is a manageable trade-off for the substantial gains in overall predictive power. Future work will explore adapting this VMD-ML framework to other asset classes and domains, and incorporating exogenous macroeconomic predictors to further refine its forecasting capabilities, solidifying its value as a versatile tool in the data science arsenal.

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